Assignment 6

CSS Flexbox and Grid — HTML/CSS Code Submission

Q1. Personal Portfolio Webpage (Flexbox)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Portfolio</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header class="site-header">
    <h1 class="logo">Aarav Sharma</h1>
    <nav class="nav-links">
      <a href="#about">About</a>
      <a href="#work">Work</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <main class="main-content">
    <section class="about" id="about">
      <h2>About Me</h2>
      <p>
        Hi, I'm Aarav — a second year student learning web development.
        I enjoy building small projects and figuring out how layouts work
        under the hood. Right now I'm getting comfortable with Flexbox and Grid.
      </p>
    </section>

    <section class="image-section">
      <img src="https://via.placeholder.com/300x300" alt="profile picture">
    </section>
  </main>

  <footer class="site-footer">
    <p>&copy; 2026 Aarav Sharma. All rights reserved.</p>
  </footer>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, Helvetica, sans-serif;
  color: #222;
}

/* header - name left, links right */
.site-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 20px 40px;
  background-color: #2c3e50;
```

```
  color: white;
}

.nav-links a {
  color: white;
  text-decoration: none;
  margin-left: 20px;
  font-size: 15px;
}

.nav-links a:hover {
  text-decoration: underline;
}

/* main - about + image side by side */
.main-content {
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 40px;
  padding: 50px 40px;
}

.about {
  flex: 1;
}

.about h2 {
  margin-bottom: 15px;
}

.about p {
  line-height: 1.6;
}

.image-section {
  flex: 1;
  display: flex;
  justify-content: center;
}

.image-section img {
  max-width: 100%;
  border-radius: 8px;
}

/* footer */
.site-footer {
  text-align: center;
  padding: 15px;
  background-color: #2c3e50;
  color: white;
}

/* stack everything below 700px */
@media (max-width: 700px) {
  .site-header {
    flex-direction: column;
    text-align: center;
  }

  .nav-links a {
    margin: 5px 10px;
  }

  .main-content {
    flex-direction: column;
    padding: 30px 20px;
  }
}
```

Q2. 4-Column Product Gallery (CSS Grid)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Product Gallery</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <h1 class="page-title">Our Products</h1>

  <main class="gallery">
    <div class="card">
      <h3>Wireless Mouse</h3>
      <p class="price">■899</p>
    </div>
    <div class="card">
      <h3>Mechanical Keyboard</h3>
      <p class="price">■3,499</p>
    </div>
    <div class="card">
      <h3>USB-C Hub</h3>
      <p class="price">■1,299</p>
    </div>
    <div class="card">
      <h3>Laptop Stand</h3>
      <p class="price">■1,799</p>
    </div>
    <div class="card">
      <h3>Webcam 1080p</h3>
      <p class="price">■2,199</p>
    </div>
    <div class="card">
      <h3>Bluetooth Speaker</h3>
      <p class="price">■2,999</p>
    </div>
    <div class="card">
      <h3>Noise Cancelling Headphones</h3>
      <p class="price">■5,499</p>
    </div>
    <div class="card">
      <h3>Portable SSD 1TB</h3>
      <p class="price">■6,999</p>
    </div>
  </main>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background-color: #f5f5f5;
  padding: 30px;
}

.page-title {
  text-align: center;
  margin-bottom: 30px;
  color: #333;
}
```

```
.gallery {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 20px;
}

.card {
  background: white;
  border-radius: 6px;
  padding: 20px;
  text-align: center;
  box-shadow: 0 2px 5px rgba(0,0,0,0.1);
  height: 150px;               /* keeps all cards same height */
  display: flex;
  flex-direction: column;
  justify-content: center;
  transition: transform 0.3s;
}

.card:hover {
  transform: scale(1.05);
}

.card h3 {
  margin-bottom: 10px;
  font-size: 17px;
}

.price {
  color: #2c7a2c;
  font-weight: bold;
}

/* tablet */
@media (max-width: 900px) {
  .gallery {
    grid-template-columns: repeat(2, 1fr);
  }
}

/* mobile */
@media (max-width: 500px) {
  .gallery {
    grid-template-columns: 1fr;
  }
}
```

Q3. 3-Section Homepage (Flexbox)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Homepage</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <nav>
      <a href="#">Home</a>
      <a href="#">Services</a>
      <a href="#">Projects</a>
      <a href="#">Contact</a>
    </nav>
  </header>

  <main>
    <div class="box">
      <h2>Services</h2>
      <p>We build websites, apps, and everything in between.</p>
    </div>
    <div class="box">
      <h2>Projects</h2>
      <p>A showcase of things we've worked on recently.</p>
    </div>
    <div class="box">
      <h2>Contact</h2>
      <p>Reach out to us anytime, we'd love to hear from you.</p>
    </div>
  </main>

  <footer>
    <p>&copy; 2026 My Company</p>
  </footer>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  display: flex;
  flex-direction: column;
  min-height: 100vh;
}

header {
  background-color: #333;
  padding: 15px 0;
}

header nav {
  display: flex;
  justify-content: space-around;
}

header nav a {
  color: white;
  text-decoration: none;
```

```
  font-size: 16px;
}

header nav a:hover {
  color: #ffd700;
}

main {
  flex: 1;
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 25px;
  padding: 50px 20px;
  flex-wrap: wrap;
}

.box {
  background-color: #f0f0f0;
  border: 1px solid #ddd;
  border-radius: 6px;
  padding: 30px;
  width: 260px;
  text-align: center;
  transition: transform 0.3s;
}

.box:hover {
  transform: rotate(2deg);
}

footer {
  background-color: #333;
  color: white;
  text-align: center;
  padding: 15px;
  width: 100%;
}
```

Q4. Dashboard Layout (CSS Grid)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dashboard</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="dashboard">
    <header class="dash-header">Admin Dashboard</header>

    <aside class="sidebar">
      <nav>
        <a href="#">Overview</a>
        <a href="#">Analytics</a>
        <a href="#">Reports</a>
        <a href="#">Settings</a>
      </nav>
    </aside>

    <main class="content">
      <div class="widget widget-wide">Sales Overview (spans 2 columns)</div>
      <div class="widget">Total Users</div>
      <div class="widget">Revenue</div>
      <div class="widget">Active Sessions</div>
    </main>
  </div>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
}

.dashboard {
  display: grid;
  grid-template-columns: 200px 1fr;
  grid-template-rows: auto 1fr;
  grid-template-areas:
    "header header"
    "sidebar content";
  min-height: 100vh;
}

.dash-header {
  grid-area: header;
  background-color: #1f2937;
  color: white;
  padding: 20px;
  font-size: 22px;
}

.sidebar {
  grid-area: sidebar;
  background-color: #374151;
  padding: 20px;
}
.sidebar nav {
```

```
  display: flex;
  flex-direction: column;
  gap: 15px;
}

.sidebar nav a {
  color: white;
  text-decoration: none;
}

.sidebar nav a:hover {
  color: #ffd700;
}

.content {
  grid-area: content;
  display: grid;
  grid-template-columns: 1fr 1fr;
  grid-gap: 20px;
  padding: 20px;
}

.widget {
  background-color: #f3f4f6;
  border-radius: 6px;
  padding: 30px;
  text-align: center;
  font-weight: bold;
  box-shadow: 0 2px 4px rgba(0,0,0,0.1);
  transition: transform 0.3s, box-shadow 0.3s;
}

.widget:hover {
  transform: scale(1.03);
  box-shadow: 0 6px 14px rgba(0,0,0,0.2);
}

.widget-wide {
  grid-column: span 2;
}
```

Q5. Pricing Section (Flexbox)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Pricing</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <h1 class="page-title">Choose Your Plan</h1>

  <div class="pricing-container">
    <div class="plan">
      <h2>Basic</h2>
      <p class="cost">■199/mo</p>
      <ul>
        <li>5 Projects</li>
        <li>Basic Support</li>
        <li>1 GB Storage</li>
      </ul>
    </div>

    <div class="plan featured">
      <h2>Pro</h2>
      <p class="cost">■499/mo</p>
      <ul>
        <li>Unlimited Projects</li>
        <li>Priority Support</li>
        <li>50 GB Storage</li>
      </ul>
    </div>

    <div class="plan">
      <h2>Enterprise</h2>
      <p class="cost">■999/mo</p>
      <ul>
        <li>Unlimited Everything</li>
        <li>24/7 Support</li>
        <li>500 GB Storage</li>
      </ul>
    </div>
  </div>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  background-color: #fafafa;
  padding: 40px 20px;
}

.page-title {
  text-align: center;
  margin-bottom: 40px;
}

.pricing-container {
  display: flex;
  justify-content: center;
```

```
  align-items: center;
  gap: 25px;
  flex-wrap: wrap;
}

.plan {
  background: white;
  border: 1px solid #ddd;
  border-radius: 8px;
  padding: 30px;
  width: 250px;
  text-align: center;
  transition: transform 0.3s;
  animation: fadeIn 0.8s ease-in;
}

.plan:hover {
  transform: translateY(-10px);
}

.plan.featured {
  background-color: #2c3e50;
  color: white;
  transform: scale(1.1);
  padding: 40px 30px;
}

.plan.featured:hover {
  transform: scale(1.1) translateY(-10px);
}

.cost {
  font-size: 26px;
  margin: 15px 0;
  font-weight: bold;
}

.plan ul {
  list-style: none;
  margin-top: 15px;
}

.plan ul li {
  padding: 6px 0;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(20px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

@media (max-width: 750px) {
  .pricing-container {
    flex-direction: column;
  }

  .plan.featured {
    transform: none;
  }

  .plan.featured:hover {
    transform: translateY(-10px);
  }
}
```

Q6. Responsive Blog Layout (CSS Grid)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Blog</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="blog-layout">
    <header class="blog-header">
      <h1>My Little Blog</h1>
      <nav>
        <a href="#">Home</a>
        <a href="#">Archive</a>
        <a href="#">About</a>
      </nav>
    </header>

    <aside class="blog-sidebar">
      <h3>Categories</h3>
      <ul>
        <li>Technology</li>
        <li>Travel</li>
        <li>Food</li>
        <li>Lifestyle</li>
      </ul>
    </aside>

    <main class="blog-main">
      <article class="post">
        <h2>Getting Started with CSS Grid</h2>
        <p>Grid makes complex layouts a lot easier once you get the hang of it...</p>
      </article>
      <article class="post">
        <h2>Why I Switched to Flexbox</h2>
        <p>For one-dimensional layouts, flexbox is honestly hard to beat...</p>
      </article>
    </main>

    <footer class="blog-footer">
      <p>Contact us: hello@myblog.com</p>
    </footer>
  </div>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Georgia, serif;
}

.blog-layout {
  display: grid;
  grid-template-columns: 220px 1fr;
  grid-template-rows: auto 1fr auto;
  grid-template-areas:
    "header header"
    "sidebar main"
    "footer footer";
```

```
  gap: 20px;
  min-height: 100vh;
  padding: 20px;
}

.blog-header {
  grid-area: header;
  display: flex;
  justify-content: space-between;
  align-items: center;
  background-color: #ffe8d6;
  padding: 20px;
  border-radius: 6px;
}

.blog-header nav a {
  margin-left: 15px;
  text-decoration: none;
  color: #333;
}

.blog-sidebar {
  grid-area: sidebar;
  background-color: #f0f0f0;
  padding: 20px;
  border-radius: 6px;
}

.blog-sidebar ul {
  list-style: none;
  margin-top: 10px;
}

.blog-sidebar ul li {
  padding: 8px 0;
  border-bottom: 1px solid #ddd;
}

.blog-main {
  grid-area: main;
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 20px;
}

.post {
  background: white;
  border: 1px solid #ddd;
  border-radius: 6px;
  padding: 20px;
}

.post h2 {
  margin-bottom: 10px;
}

.blog-footer {
  grid-area: footer;
  text-align: center;
  background-color: #ffe8d6;
  padding: 15px;
  border-radius: 6px;
}

@media (max-width: 800px) {
  .blog-layout {
    grid-template-columns: 1fr;
    grid-template-areas:
      "header"
      "main"
      "sidebar"
      "footer";
  }
  .blog-main {
```

```
    grid-template-columns: 1fr;
  }
}
```

Q7. Loading Animation (Div + Keyframes)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Loading Animation</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="loader-wrapper">
    <div class="dot"></div>
    <div class="dot"></div>
    <div class="dot"></div>
  </div>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  background-color: #111;
}

.loader-wrapper {
  display: flex;
  gap: 12px;
}

.dot {
  width: 18px;
  height: 18px;
  border-radius: 50%;
  background-color: #00c9a7;
  animation: bounce 0.6s infinite ease-in-out;
}

.dot:nth-child(1) {
  animation-delay: 0s;
}

.dot:nth-child(2) {
  animation-delay: 0.2s;
}

.dot:nth-child(3) {
  animation-delay: 0.4s;
}

@keyframes bounce {
  0%, 100% {
    transform: translateY(0);
  }
  50% {
    transform: translateY(-20px);
  }
}
```

Q8. Photo Gallery with Hover Zoom (CSS Grid)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Photo Gallery</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <h1 class="page-title">Photo Gallery</h1>

  <div class="gallery">
    <figure>
      <img src="https://via.placeholder.com/300x200?text=Photo+1" alt="photo 1">
      <figcaption>Mountains at sunrise</figcaption>
    </figure>
    <figure>
      <img src="https://via.placeholder.com/300x200?text=Photo+2" alt="photo 2">
      <figcaption>City skyline</figcaption>
    </figure>
    <figure>
      <img src="https://via.placeholder.com/300x200?text=Photo+3" alt="photo 3">
      <figcaption>Beach evening</figcaption>
    </figure>
    <figure>
      <img src="https://via.placeholder.com/300x200?text=Photo+4" alt="photo 4">
      <figcaption>Forest trail</figcaption>
    </figure>
    <figure>
      <img src="https://via.placeholder.com/300x200?text=Photo+5" alt="photo 5">
      <figcaption>Old town street</figcaption>
    </figure>
    <figure>
      <img src="https://via.placeholder.com/300x200?text=Photo+6" alt="photo 6">
      <figcaption>Desert dunes</figcaption>
    </figure>
  </div>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
  padding: 30px;
  background-color: #fff;
}

.page-title {
  text-align: center;
  margin-bottom: 25px;
}

.gallery {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 15px;
}

.gallery figure {
  margin: 0;
```

```
  overflow: hidden;
  border-radius: 6px;
  background-color: #f0f0f0;
}

.gallery img {
  width: 100%;
  height: 200px;
  object-fit: cover;
  display: block;
  transition: transform 0.5s ease;
}

.gallery img:hover {
  transform: scale(1.1);
}

.gallery figcaption {
  text-align: center;
  padding: 8px;
  font-size: 14px;
  color: #444;
}
```

Q9. Card Flip Animation (CSS Transforms)

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Card Flip</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="scene">
    <div class="card">
      <div class="card-face card-front">
        <img src="https://via.placeholder.com/200x120?text=Front" alt="front">
        <h3>Flip Me</h3>
      </div>
      <div class="card-face card-back">
        <p>This is the back of the card. Hover away and it flips back to the front side.</p>
      </div>
    </div>
  </div>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  background-color: #eee;
  font-family: Arial, sans-serif;
}

.scene {
  perspective: 1000px;
}

.card {
  width: 220px;
  height: 280px;
  position: relative;
  transform-style: preserve-3d;
  transition: transform 0.7s;
}

.scene:hover .card {
  transform: rotateY(180deg);
}

.card-face {
  position: absolute;
  width: 100%;
  height: 100%;
  backface-visibility: hidden;
  border-radius: 10px;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  padding: 15px;
```

```
  text-align: center;
}

.card-front {
  background-color: white;
  border: 1px solid #ccc;
}

.card-front img {
  max-width: 100%;
  margin-bottom: 10px;
  border-radius: 6px;
}

.card-back {
  background-color: #2c3e50;
  color: white;
  transform: rotateY(180deg);
}
```

Q10. Combined Flexbox + Grid Layout

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Combined Layout</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header class="top-header">
    <h1 class="logo">MyBrand</h1>
    <nav>
      <a href="#">Home</a>
      <a href="#">Blog</a>
      <a href="#">Contact</a>
    </nav>
  </header>

  <main class="page-grid">
    <article class="article">
      <h2 class="fade-title">Building Layouts with Flexbox and Grid</h2>
      <p>Combining both tools gives you the best of both worlds — Grid for
        overall page structure, and Flexbox for aligning smaller groups
        of items inside a section.</p>

      <div class="article-boxes">
        <div class="mini-box">Tip 1</div>
        <div class="mini-box">Tip 2</div>
        <div class="mini-box">Tip 3</div>
      </div>
    </article>

    <aside class="sidebar">
      <h3>Recent Posts</h3>
      <ul>
        <li>CSS Grid Basics</li>
        <li>Flexbox Cheatsheet</li>
        <li>Responsive Design Tips</li>
      </ul>
    </aside>
  </main>

  <footer class="bottom-footer">
    <p>&copy; 2026 MyBrand — All rights reserved</p>
  </footer>

</body>
</html>
```

style.css

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: Arial, sans-serif;
}

/* header - flexbox */
.top-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 20px 40px;
  background-color: #1e3a5f;
```

```
  color: white;
}

.top-header nav a {
  color: white;
  text-decoration: none;
  margin-left: 20px;
}

/* main - grid: article + sidebar */
.page-grid {
  display: grid;
  grid-template-columns: 2fr 1fr;
  gap: 25px;
  padding: 30px 40px;
}

.article {
  background-color: #f9f9f9;
  border-radius: 6px;
  padding: 25px;
}

.fade-title {
  animation: fadeUp 1s ease-out;
  margin-bottom: 15px;
}

.article p {
  line-height: 1.6;
  margin-bottom: 20px;
}

/* boxes inside article aligned with flexbox */
.article-boxes {
  display: flex;
  gap: 15px;
}

.mini-box {
  flex: 1;
  background-color: #1e3a5f;
  color: white;
  padding: 15px;
  text-align: center;
  border-radius: 4px;
}

.sidebar {
  background-color: #eef2f7;
  border-radius: 6px;
  padding: 20px;
}

.sidebar ul {
  list-style: none;
  margin-top: 10px;
}

.sidebar ul li {
  padding: 8px 0;
  border-bottom: 1px solid #ddd;
}

.bottom-footer {
  text-align: center;
  background-color: #1e3a5f;
  color: white;
  padding: 15px;
}

@keyframes fadeUp {
  from {
    opacity: 0;
```

```
    transform: translateY(15px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

/* mobile - single column */
@media (max-width: 700px) {
  .top-header {
    flex-direction: column;
    text-align: center;
  }

  .top-header nav a {
    margin: 5px 10px;
  }

  .page-grid {
    grid-template-columns: 1fr;
    padding: 20px;
  }

  .article-boxes {
    flex-direction: column;
  }
}
```